

Norm Emergence Report: Resistance

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1 Introduction

Overall the student report has some useful ideas which can be added fairly quickly to the current model. The report correctly identifies that the gossip mechanism is central to norm formation or subgroup resistance. However, they treat it as the source of these behaviours and not as an emergent structure in a system predisposed to them. Most of the decisions regarding what to keep or remove from the report follow from this observation.

2 Additions From the Report

Gossip

The student report identifies gossip as the central mechanism that connects agents and would drive resistance, which is clear from the effect of gossip on the current simulations. However, I believe they misclassified its role in the emergence of a resistant subgroup.

Namely, the student report suggested a modified gossip algorithm such that the unhappy subgroup has a separate gossip network on top of the collective network for the entire group. I believe this should be an emergent from the difference in reward structure between the two groups and not artificially created.

However, the student report correctly identified that in the current simulation, there is no way for such a gossip network to form. This is because each gossip update is of the form:

$$s_i(k, t + 1) = \frac{1}{|\mathcal{A}_p(t)|} \sum_{j \in \mathcal{A}_p(t)} s_j(k, t) + (v_i(k, t) - v_i(k, t - 1))$$

where $s_i(k, t)$ is i 's reputation estimate for k at time t , $\mathcal{A}_p(t)$ is the set of participating actors at time t and $v_i(k, t)$ is i 's estimate of its personal benefit from k at time t .

Now as $t \rightarrow \infty$ theoretically we find that

$$s_i(k, t) \approx s_j(k, t)$$

as the consensus term is identical for all participants and $v_i(k, t)$ plateaus as $t \rightarrow \infty$ which implies $v_i(k, t + 1) - v_i(k, t) \rightarrow 0$.

This can be verified empirically for sufficiently large t close to norm convergence. This drives the adoption of the current leader's norm and prevents any resistance from forming.

Similarity

Another proposed addition from the student report can resolve the gossip mechanism by naturally creating the separated gossip networks for the total group and unhappy subgroup. This addition is the similarity score, defined for a state $s \in S$ as

$$\text{sim}_t(i, j, s) = \begin{cases} \text{sim}_{t-1}(i, j, s) \cdot (1 - \beta) + \beta \frac{1}{|K_s(t)|} \sum_{k \in K_s(t)} \left(u_i(t, s, \pi_k(s)) - u_j(t, s, \pi_k(s)) \right)^2 & t > 0 \\ 0 & t = 0 \end{cases}$$

where $K_s(t)$ is the set of actors acting in state s at time t . if $K_s(t) = \emptyset$, then there is no update. for a fixed rate constant $\beta \in (0, 1)$ then

$$\text{sim}_t(i, j) = \frac{1}{|S|} \sum_{s \in S} \text{sim}_t(i, j, s)$$

then there is a fixed constant C such that agents i and j could gossip at time t only if

$$\text{sim}_t(i, j) < C$$

and an intuitive hypothesis would be that overtime unhappy agents will have a low score with each other and a high score with other agents and this would naturally fragment the gossip network into two subnetworks.

Additionally, a utility matrix is computed at each time step of the simulation, so implementing such a similarity score would require minimal tweaking to the current model.

Modelling the Unhappy Agents

The report proposes to model a subset of unhappy agents as a subgroup G_r of the total agents G such that on a subset of the states $S_r \subseteq S$ the dominant schema σ_d is significantly inferior to the potential resistant schema σ_r , i.e.

$$u_i(s, \sigma_r) \gg u_i(s, \sigma_d) \quad \forall i \in G_r, \forall s \in S_r$$

and this is a reasonable gap to drive the formation of a resistant subgroup that fragments from the dominant group.

However, the report did not address how agents in G_r will discover the imbalance in personal utility under the dominant schema. This is because in the current simulation, the personal utility estimate of an agent is unchanged when the agent is optimizing status or reputation. Additionally, for an agent that has adopted the dominant schema as a follower of the current leader, it only acts under the policy of the leader without the chance to sample from its own personal utility weights which is how its personal utility estimate could rise above its reputation estimate to detach it from the group.

To fix these two issues, the personal utility estimate of each agent must be kept current and their personal utility weights must not be overridden by the leader weights.

For the agents to actually occasionally witness the schema σ_r I propose an extension of the report model by adding another rate μ_{dev} where $\theta(\mu_{\text{dev}})$ gives the probability that a given agent, regardless of whether they optimize personal utility, reputation or status, will instead choose to optimize personal utility at that given time step. This deviation mechanism will allow unhappy agents to explore other schemas which in turn can facilitate the formation of a resistant subgroup.

Note that for μ_{dev} to have any affect the current model has to be tweaked, as at the moment an agent i that optimizes reputation copies its leader's weights and does not keep track of its previous personal utility weights.

Another implementation detail, if two norms emerge in the simulation multiple functions in the model will fail silently as they assume a single norm. These include `paper_welfare` which takes a single `leader_id` and the `leader metric` which takes a global `argmax` of follower counts. Therefore, migrating the code from a single norm simulation to one with two will require a lot of attention to which functionality implicitly assumes a single norm.

Attention Allocation for Participant Rate

The student paper proposes a new way to model the interaction μ_i . Namely

$$\{\mu_i(s)\}_{s \in S} = \operatorname{argmax}_{\mu_i} \sum_{s \in S} \theta_\alpha(\mu_{\min} + \mu_i(s)) \cdot u_i(s, \pi_i(s))$$

subject to the constraints

$$\sum_{s \in S} (\mu_{\min} + \mu_i(s)) \leq M \quad 0 \leq \mu_i(s)$$

where the agent i 's interaction rate becomes $\mu_{\text{total}} = \mu_{\min} + \mu_i$ here M is a budget on the total attention an actor i can allocate to different situations and $\mu_{\min} > 0$ to prevent degenerate behaviour. Under this model it is possible for an agent to selectively focus their attention on states that benefit them.

Additionally θ_α is a generalization of θ which is defined as

$$\theta_\alpha(\mu) = 1 - \exp(-\alpha\mu)$$

although up to scaling of μ and M , α is redundant so we take it to be equal to 1.

This optimization problem has a closed form solution from the KKT conditions.

Specifically there emerges a partition of the states $F \sqcup P = S$ called the free and pinned states where for all $s \in P$

$$\mu_i(s)^* = 0$$

then for $s \in F$

$$\mu_i(s)^* = \frac{M}{|F|} - \left(\mu_{\min} + \frac{|P|}{|F|} \mu_{\min} \right) + \left(\log u_i(s, \pi_i(s)) - \frac{1}{|F|} \sum_{t \in F} \log u_i(t, \pi_i(t)) \right)$$

which will be positive.

then for $s \in G_r$ still following the dominant schema σ_d if for a moment they deviate through μ_{dev} and act on personal utility they will update $\mu_i(s)$ based on σ_r and this will drive $\mu_i(s)$ up for $s \in G_r$ and down for $s \in G \setminus G_r$. Hence this attention mechanism should amplify deviation in the subgroup into cohesion of the subgroup.

The report did not specify where this model for μ_i would be used in the simulation, but I propose that it replaces the current model for $\mu_{i,a}$.

An intuitive hypothesis would be that as the system progresses and the resistant subgroup forms, a given member $s \in G_r$ its free states will correspond roughly to S_r and its pinned states to $S \setminus S_r$. And the opposite effect would be present for $G \setminus G_r$. This would reinforce a fragmented gossip network when combined with the similarity neighborhood.

In terms of implementation, the report does not give any method for how this interaction rate would be updated during simulation. However, I designed a method that mirrors the approach for the current actor interaction rate optimization.

For each agent i , given some initialization of $\mu_{i,a}(s, 0)$ for all $s \in S$ that satisfies

$$\sum_{s \in S} (\mu_{i,a}(s) + \mu_{\min}) \leq M \quad \forall s \in S, \mu_{i,a}(s) \geq 0$$

at time t the above optimization problem is solved and $\mu_{i,a}(s, t)^*$ are found for $s \in S$. Then there is a fixed rate constant $\eta(t) \in (0, 1)$ such that the rate is updated to

$$\mu_{i,a}(s, t+1) = (1 - \eta(t))\mu_{i,a}(s, t) + \eta(t) \cdot \mu_{i,a}(s, t)^*$$

and this clearly satisfies the constraints as $\{\mu_{i,a}(s, t)\}_{s \in S}$ and $\{\mu_{i,a}(s, t)^*\}_{s \in S}$ both do.

$\eta(t)$ should satisfy the same constraints as the rate variables in the learning norms paper. Namely

$$\begin{aligned} \sum_{n=0}^{\infty} \eta(t)^2 &< \infty \\ \sum_{n=0}^{\infty} \eta(t) &= \infty \end{aligned}$$

Now to actually calculate $\mu_i(s)^*$ in practice F, P have to be computed. To do this we sort S as $\{s_1, \dots, s_N\}$ in increasing order with respect to $u_i(s_j, \pi_i(s_j))$. It is clear that P is of the form $\{s_j : 0 < j \leq k\}$ for some $k \in \{0, \dots, N+1\}$ Hence we can define

$$L_k = \sum_{j \leq k} u_i(s_j, \pi_i(s_j))\theta(\mu_{\min}) + \sum_{j > k} u_i(s_j, \pi_i(s_j))\theta(\mu_{i,a}(s_j, t) + \mu_{\min})$$

then $P = \{s_j : 0 < j \leq k\}$ for

$$k = \operatorname{argmax}\{L_k : 0 \leq k \leq N+1\}$$

for every agent this has worst case complexity $\mathcal{O}(|S| \log |S|)$ and since this is done for each agent, every participant interaction rate update would have worst case complexity $\mathcal{O}(|G||S| \log |S|)$ which is reasonable for moderately sized simulations.

The only implementation issue would be that this attention mechanism requires some significant changes to the current code and could easily cause silent failures. Hence the process of migrating from the current actor interaction rate to this attention-based one has to be done carefully.

Note that I took $\mu_{\min}$ as uniform across all states to simplify the optimization, although in the report they allow $\mu_{\min}$ to vary across states which is a future extension to the above model.

Additionally this could be extended even further to $\mu_{i,p}$, the only difference is that instead of weighting each state term by an agents own reward $u_i(s, \pi_i(s))$ it would be better to weight it by

$$H_i(s) = \frac{1}{|G|} \sum_{k \in G} u_i(s, \pi_k(s))$$

then solve the optimization problem

$$\{\mu_{i,p}(s)\}_{s \in S} = \operatorname{argmax}_{\mu_i} \sum_{s \in S} \theta_\alpha(\mu_{\min} + \mu_{i,p}(s)) \cdot H_i(s)$$

subject to the constraints

$$\sum_{s \in S} (\mu_{\min} + \mu_{i,p}(s)) \leq M \quad 0 \leq \mu_i(s)$$

and the closed form solution can be found using the same method as the previous problem only replacing $u_i(s, \pi_i(s))$ with $H_i(s)$.

Reward Table Structure

The report proposes a reward table structure to model σ_r and σ_d that is intuitive and produces S_r and G_r as discussed above. Namely for states $s \in S \setminus S_r$ agent rewards can be sampled under one mean and standard deviation. However for $s \in S_r$, actions associated to the resistant schema σ_r will a reward distribution given by

$$u_i(s, \sigma_r) \sim \mathcal{N}(m_r^{G_r}, \sigma_r^{G_r^2}) \quad \forall i \in G_r$$

and

$$u_i(s, \sigma_r) \sim \mathcal{N}(m_r^G, \sigma_r^{G^2}) \quad \forall i \in G \setminus G_r$$

then for the dominant schema σ_d :

$$u_i(s, \sigma_d) \sim \mathcal{N}(m_d^{G_r}, \sigma_d^{G_r^2}) \quad \forall i \in G_r$$

and

$$u_i(s, \sigma_d) \sim \mathcal{N}(m_d^G, \sigma_d^{G^2}) \quad \forall i \in G \setminus G_r$$

where the marginalized group is created by enforcing

$$m_r^{G_r} \gg m_d^{G_r} \quad m_d^G \gg m_r^G$$

this sort of reward table can be built on top of the current reward table structure in the simulation without many changes.

3 What Will Not be Used

Multiple Gossip Networks

The report proposed a manually fragmented gossip network, which I have decided to scrap in favour of using the above similarity neighborhoods to produce a fragmented gossip network naturally. My approach is aligned with the social reward economies paper which treats gossip networks as emergent.

Separate Gammas

The model proposes several γ values that modulate the weight of an agent i 's reputation in G versus G_r for $i \in G_r$ versus $i \in G \setminus G_r$. This is an artificial addition as it indexes γ by group membership to create a gap in reputation for different groups. However, this difference is precisely what should emerge from the fragmentation of the gossip network mentioned above.

Experiment Setup

All of the experiment set up in the report makes sense, any parameters which I have removed above such as the different γ values will be removed. However, one point where there the experiment should have a different setup is at initialization, namely the current experiment seeds all agents with the dominant norm except those in the unhappy group G_r which are given σ_r . However this artificially forms the resistant subgroup and only studies whether or not it survives over time. This is a separate question from whether or not this subgroup will form on its own. Before the changes to personal utility tracking from above are implemented, it would have been impossible for the resistant subgroup to form and seeding the resistant subgroup with the resistant schema would have been necessary for any behaviour to emerge. However, with my proposed changes to personal utility tracking and follower deviation, the subgroup formation can happen naturally.

Parameter Variation in Experiments

The report proposes that each of its experiment parameters are varied one at a time for each experiment run, this is not optimal as it assumes the effect of each parameter is additive. Since I have already cut down the number of parameters in the simulation I propose instead to run a grid of different parameter values. Even if this does not mean all parameters are in the grid, even a experiment with a grid of 2 or 3 parameters will give richer results than an experiment that only varies one.

In terms of specific parameters, the fixed parameters the report proposed which will still be present are

- 1 N: the number of agents
- 2 S: the set of situations
- 3 M: the total attention budget
- 4 γ : the reputation scaling factor
- 5 κ : the status scaling factor
- 6 Normal distribution parameters for reward tables

and the variable parameters are

- 1 The distribution of $\mu_{i,a}(s)$.
- 2 The distribution of $\mu_{i,p}(s)$.
- 3 The distribution of $\mu_{\min}(s)$ (if taken to be non-uniform) versus total utility for each situation.

Additionally some parameters not mentioned in the report that would be added are

- 1 C: the similarity score threshold to enable gossip.
- 2 μ_{dev} : the rate modulating the chance of follower deviation.